



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD 6.GR.4

Surface Area Using Nets

Lesson 10-3 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can find the surface area of a solid by using its net.

Language Objective: I can explain my work using the words surface area, net, face, and two-dimensional.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area

Net

Face

Two-dimensional

Edge



Tap any word to see what it means and a picture.

1 The total area of all the faces of a solid is its .

2 A flat pattern that folds up into a solid is a .

3 A flat surface of a solid figure is a .

4 A flat shape with length and width but no thickness is .

5 A line segment where two faces meet is an .

Reflect — Exit Ticket

A rectangular prism has $l = 9$ in, $w = 5$ in, $h = 3$ in. What is the surface area?

- A. 174 in^2
- B. 135 in^2
- C. 174 in^3
- D. 87 in^2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Net
3. **Notes 3:** Face
4. **Notes 4:** Two-dimensional
5. **Notes 5:** Edge
6. **Watch for this mistake:** *A common mistake in Surface Area Using Nets is skipping the key idea: "A box has 3 pairs of matching faces, so add up all 6 faces to get the surface area." — always check your work against this rule before you submit.*
7. **Try It 1:** A. 6 — *A rectangular prism has 6 faces — top, bottom, front, back, left, and right — so its net shows 6 faces.*
8. **Try It 2:** A. 52 in^2 — *From the net: $SA = 2(2 \times 3) + 2(2 \times 4) + 2(3 \times 4) = 12 + 16 + 24 = 52 \text{ in}^2$.*
9. **Exit Ticket:** A. 174 in^2 — *$SA = 2(9 \times 5) + 2(9 \times 3) + 2(5 \times 3) = 90 + 54 + 30 = 174 \text{ in}^2$. Surface area uses square units (in^2).*